Prodigy Anesthesia Quick Review

DIC

What is disseminated intravascular coagulation (DIC) and how does it occur?	DIC is a condition that arises when the coagulation system becomes activated over such as widespread area that small clots begin to occlude the small peripheral blood vessels throughout the body. Excessive fibrinolysis to compensate for the consumption of coagulation factors during the clotting process also contributes to the progression of DIC. It commonly occurs due to massive trauma or the presence of a large amount of dying tissue that releases tissue factor into the bloodstream. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 459.*
Why does DIC often result in bleeding?	Because the coagulation pathway is activated throughout the body on a massive scale, the clotting factors can become exhausted. As more thromboses form, the coagulation factors become so diminished that spontaneous bleeding can occur in other parts of the body. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 424.*
What are some known causes of DIC?	Sepsis, leukemia, cancer, and trauma. Placenta previa, abruption, amniotic fluid embolism, or the presence of a dead fetus late in pregnancy can also result in DIC. Prolonged surgery and surgery for certain procedures such as prostatectomy for carcinoma, liver transplantation, and coronary bypass surgery are associated with significant bleeding abnormalities. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 897.*
How can DIC lead to organ failure?	The widespread bleeding and thrombosis can cut off the blood supply to tissues and organs, resulting in ischemia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 894.*
What laboratory values are decreased in patients with DIC?	Fibrinogen, platelet count, factor V, and factor VIII will all be decreased in patients with DIC. *Ferri FF. Ferri's Best Test. 3rd ed. Philadelphia, PA: Elsevier Saunders; 2015: 247.*
What laboratory values are increased in patients with DIC?	The PT, PTT, thromboplastin time, D-dimer, and fibrin degradation products are all elevated in patients with acute DIC. *Ferri FF. Ferri's Best Test. 3rd ed. Philadelphia, PA: Elsevier Saunders; 2015: 247.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

DIC

How is sepsis related to DIC?	Endotoxins released from the bacteria can activate the coagulation pathway. The microthromboses that form can decrease the oxygen supply to the tissues and exacerbate shock. DIC is one of the reasons that the mortality rate for septic shock is as high as 85%. Although DIC may occur in relation to gram-positive or even viral or fungal infections, it is most strongly associated with gram-negative sepsis. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 459.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.